

Data Preparation Session 1

Exercises

Exercise 1.0

If not done yet, open a new .R script in R Studio. Name your script by the schema of FAMILYNAME_Firstname_Session1.R (and please use your own name, I don't want to receive 50 files named exactly as "FAMILYNAME_Firstname_Session1.R"). You can save and reuse this script for all exercises in this session.

Exercise 1.1

- a) Put the numbers 31 to 60 in a vector named P and in a matrix with 6 rows and 5 columns named Q.
- b) Create a matrix Qt where you fill first the first row, then the second, and so on.

Exercise 1.2

- a) Construct three normal random vectors of 100 length (in other words, each vector contains 100 random numbers from a standard normal distribution). Use the rnorm() function. Name these vectors x1, x2 and x3.
- b) Make a data frame called t with three columns (called a, b and c) containing respectively x1, x1+x2 and x1+x2+x3.
- c) Call the following functions for this data frame: plot(t) and sd(t). Can you understand the results? (Short comment starting with # is welcome.)
- d) Modify sd(t) to get the expected results about the three variables in data frame t.

Exercise 1.3

Create a vector using a sequence going from 21 to 120. Give it the name "a".
Define b as the length of a.
Define d the 5th element of the vector a.
Define f the vector containing the elements from the 2nd until the 6th.
Define g the vector containing the 1st, 3rd and 7th elements of a.
Create a vector, call it h, containing the sequence of values of "a" from 1 until 100 every 4 observations.
Define i the vector containing the elements bigger than 24 and smaller than 29.
Create a matrix (give it the name l) with 25 rows and 4 columns containing the element of the vector a.
Define m the vector containing the elements of the second column of l.
Define n the vector containing the elements of the third row of l.
Define o the vector containing the elements included from row 6 until row 12 and from column 2 until column 3 of l.

Exercise 1.4

- a) Create a vector months with values Feb, Apr, Jun, Jul, Aug. Print the vector.
- b) Sort months by using sort(). What order do we have?
- c) Create a vector with the months in the right order:

```
month_levels <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug",  
"Sep", "Oct", "Nov", "Dec")
```
- d) Create a factor mymonths using the vector months and the levels in month_levels. Print mymonths.
- e) Sort mymonths. Comparing results in b), d) and e), why are factors useful tools?